

Milestone 4 Technical Summary

Drug Review Patient Satisfaction Prototype

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Project Goal

The system predicts the satisfaction expressed in a real patient-written medication review. The input is a review plus optional drug and condition context. The output is a low, medium, or high satisfaction label, class probabilities, positively contributing evidence terms, and three similar reviews retrieved from the training corpus. The prototype analyzes patient experience only; it does not diagnose, prescribe, or recommend treatment.

Architecture and Design

The notebook loads the provided drugsComTrain_raw(1).csv and drugsComTest_raw(1).csv files. It decodes HTML entities, normalizes whitespace, fills missing context, and maps ratings 1-4 to **low**, 5-6 to **medium**, and 7-10 to **high**. Drug name, condition, and review text are concatenated into one model input.

The primary path is: **review + context -> TF-IDF -> balanced logistic regression -> label + probabilities + evidence terms**. TF-IDF is fast and interpretable, while logistic regression provides a strong transparent baseline for text classification. Class weighting reduces the tendency to ignore the smaller medium class.

A retrieval-style grounding layer reuses the trained TF-IDF space and cosine nearest neighbors. It returns source-row identifiers, similarity scores, ratings, labels, and text for three similar training reviews. These examples create an audit trail, but they are patient anecdotes rather than clinical evidence. A lightweight chatbot-style function exposes this pipeline and refuses medication, dosage, or treatment requests.

Input and Output Contract

Input: one patient review (required), drug name (optional), and condition (optional). **Output:** predicted satisfaction, per-class probabilities, evidence terms, similar source reviews, and a safety note. Empty inputs receive a prompt to enter a review; medical-advice requests receive a refusal and clinician referral.

Evaluation Against Baseline

The model trains on a reproducible 40,000-row sample and evaluates on 12,000 held-out rows from the separate test file. Accuracy measures overall correctness; macro-F1 weights each class equally and is essential because the class distribution is imbalanced.

System	Accuracy	Macro-F1
Majority baseline	65.9%	0.265
TF-IDF + logistic regression	73.5%	0.603

The prototype improves accuracy by **7.6 percentage points** and more than doubles macro-F1. Performance is strongest for high-satisfaction reviews and weakest for medium reviews, whose ratings often represent mixed experiences.

Grounding and Hallucination Analysis

The primary classifier does not generate free-form medical answers, so classic generative hallucination is limited. It can still misread negation, sarcasm, or mixed experiences and can present an incorrect label with high apparent confidence. Grounding is therefore provided at two levels: evidence terms identify input features that positively support the predicted class, and nearest-neighbor retrieval presents similar source reviews with row identifiers and similarity scores.

Retrieval is not evaluated as proof that the prediction is medically correct. Similarity can be driven by a drug name, condition, or repeated wording, and training reviews may themselves be inaccurate. The interface explicitly labels retrieved items as patient experiences and never as clinical citations.

Risks and Limitations

Label noise. Numeric ratings are subjective and may conflict with written text. A patient can report serious side effects but still assign a high rating because a medication worked, or assign a low rating because of cost or expectations.

Data limitations. Reviews are self-reported and may contain selection bias, incomplete histories, spelling variation, sarcasm, duplicated language, and medically inaccurate statements. The sample is not representative of all patients or conditions.

Model limitations. The medium class is small and linguistically ambiguous. Probabilities are not calibrated. Evidence terms describe model contributions, not causal explanations. Retrieval from the same training corpus can surface close duplicates and should be deduplicated in a production system.

Safety limitations. A satisfaction prediction must never be treated as evidence that a medication is safe or effective for an individual. The system must not recommend drugs, compare treatments, propose doses, or replace a clinician.

Safeguards

The prototype refuses common treatment and dosage requests, returns a medical-safety disclaimer, separates patient anecdotes from clinical evidence, and limits its response to review-satisfaction analysis. Its outputs should be used for aggregate patient-experience research, with human review for consequential decisions.

What Works and Next Steps

The notebook demonstrates the full course requirement: real input data, a reproducible input-to-output system, a baseline comparison, held-out evaluation, interpretable outputs, retrieval-style grounding, qualitative error inspection, and a safe interaction layer.

For Milestone 5, the team should (1) calibrate probabilities; (2) evaluate by condition, drug category, and review length; (3) inspect high-confidence errors and rating-text conflicts; (4) compare alternative thresholds or a binary formulation; (5) detect near-duplicate reviews before retrieval; and (6) conduct a structured human evaluation of whether evidence terms and retrieved examples are useful and non-misleading.

Reproducibility and Submission

The final notebook includes fixed random seed 42 and explicit sample sizes. Before submission, the team must add its real GitHub notebook URL, upload both CSV files to the Colab session, restart the runtime, run all cells in order, confirm that no cell errors, and export the executed notebook as PDF. Canvas should receive the GitHub URL, notebook PDF, and this two-page technical summary.